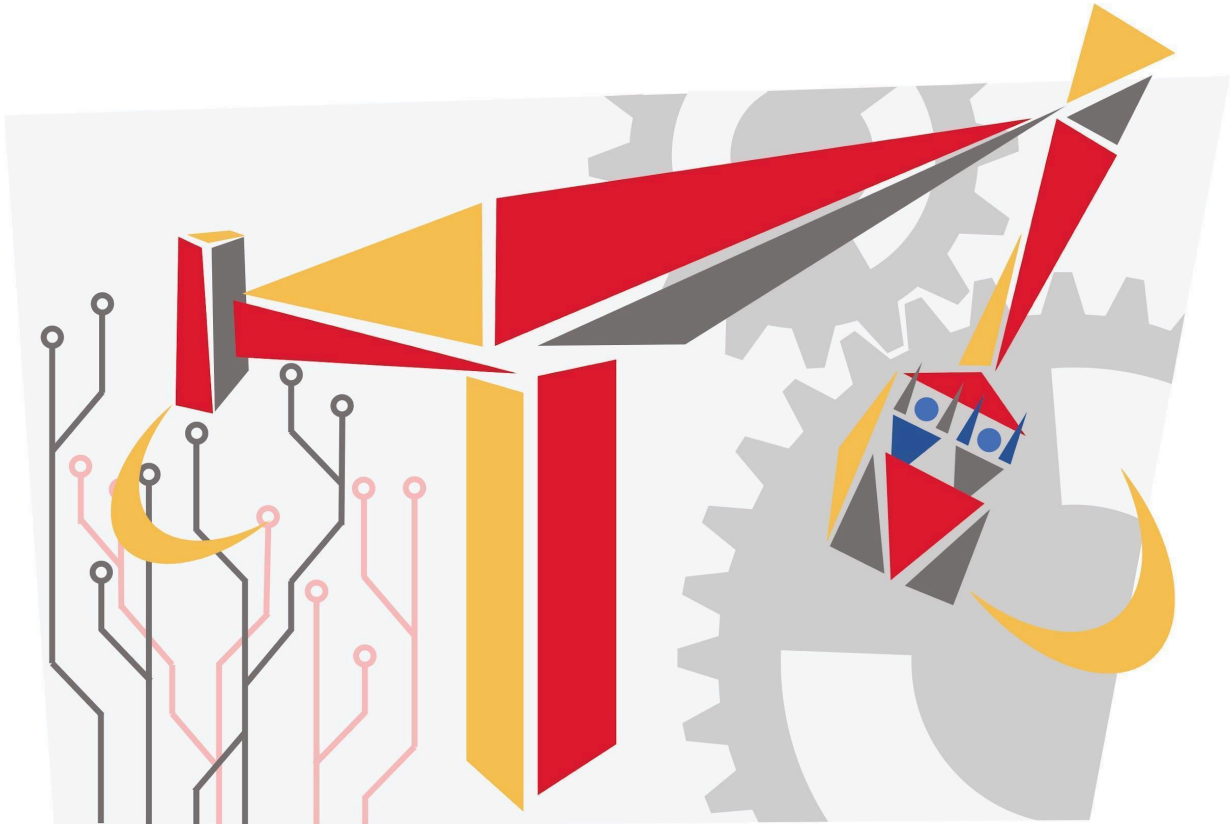


Ride Engineering Competition



ASTM Adaptation Document 2026

The Ride Engineering Competition

Purpose

The ASTM F24 body of standards has been thoroughly developed by large numbers of industry professionals over the course of many years. These standards serve to ensure the safety of Amusement Rides and Devices. It is the responsibility of an engineer to understand and comply completely to all standards applicable to their work. The ASTM Compliance portion of the Ride Engineering Competition is designed to create a simplified educational experience where teams are only responsible for compliance to one real-world standard: F2291-25b Standard Practice for Design of Amusement Rides and Devices. (2025)

F2291 is written specifically for rides carrying human riders, and has many specific values and figures that are not feasible for the context of the Ride Engineering Competition's small scale inanimate riders. Additionally, there are many requirements in the standard that are outside the scope of the competition, or are impractical or impossible for teams to comply with.

Due to these differences, this ASTM Adaptation Document shall be used to filter and adapt the F2291 Standard for the context of the Ride Engineering Competition.

How to use

When teams create their Compliance analysis, they are responsible for showing whether each requirement applies to their design, how they comply with the requirement that applies, and a rationale for each. For many requirements, this document shall serve as that rationale. The language in this document is not to be considered a standalone safety standard requirement, but rather to allow for reduction of scope for requirements that are not feasible, and allow for compliance using substitute values, as determined by the REC Planning Committee.

A team could state in the Compliance analysis for example:

ASTM F2291 Section 7.1.1.2 states: "The acquired [Acceleration] test data shall be post-processed, with a 4-pole, single pass, Butterworth low pass filter using a corner frequency (Fn) of 5 Hz.

This Requirement does not apply to our ride. The REC 2026 ASTM Adaptation Document states that Section 7.1.1.2 does not fall within the scope of the Ride Engineering Competition.

Obtaining F2291-25b

Most universities subscribe to the ASTM book of standards and students can download standards from the ASTM Compass portal. The ASTM F24 Committee has made an effort to make its standards available to all students by creating an ASTM account as a student and accessing their website [here](#). If you do not have access to these standards at your university please contact us.

Questions

This standard is complex, and we may have missed something. If you have any questions about using this document, or how any specific section applies, please contact the REC Planning committee in the by email at info@rideengineeringcompetition.org. Any updates or clarifications during the season will be communicated to all teams in the REC 2026 Discord Server.

Unless otherwise stated, Teams should assume that every requirement could apply to their ride, and should expect to document justification for each requirement that they determine does not apply.

Adaptations:

SECTION 1: Scope

This Standard Shall apply to the Ride Engineering Competition Only with the following adaptations:

SECTION 2: Referenced Documents

No References are required as part of the REC

SECTION 3: Terminology

No Change

SECTION 4: Significance and Use

No Change

SECTION 5: General Design Criteria

5.3.1.4 - See REC FIG. 1 Below

5.5.2 - See REC FIG. 1 Below

5.6.1 - Disregard retention requirement (Production still required)

5.6.2 - Omit; Confidentiality is not permitted in REC deliverables

5.7.1.4 - Reference not required; Refer to REC Rule Book for any additional Information Required

SECTION 6: Patron Restraint, Clearance Envelope, and Containment Design Criteria

6.3.1 - Refer to Rider characteristics detailed in the REC Prompt

6.4.3.7 - Redundancy not required for the locking device function

6.4.3.8 - Redundancy not required for the locking device function

6.6.3 - The minimum patron model shall be based on the rider characteristics included in the provided REC Prompt, with an additional (extended) arm reach of not less than 1.5 mm. (Replacing the 3 inches value in Fig. 3)

SECTION 7: Acceleration Limits

7.1.1 - Omit; No Test Data Required

7.1.1.1 - Omit; No Test Data Required

7.1.1.2 - Omit; No Test Data Required

7.1.1.3 - Omit; No Test Data Required

7.1.2 - Expert Review not Required

7.1.3 - See REC FIG. 1 Below

7.1.4.3 - Omit; Out of Scope

7.2.1.1 - Omit; Expert Review not Required

SECTION 8: *Loads and Strengths*

8.3 - The REC shall use a minimum 6 operational hour criteria

8.4.2 - Include this list in the safety documentation for the REC; Reference not Required

8.5.1 - Omit; Out of Scope

8.6.1 - Use upper weight limit described in the REC Rule Book

8.6.2 - Use lower weight limit described in the REC Rule Book

8.6.3 - Use upper weight limit described in the REC Rule Book

8.6.5 - Use lower weight limit described in the REC Rule Book

8.6.8 - Omit; Out of Scope

8.6.8.1 - Omit; Out of Scope

8.11.1 - Consider forces assembling / servicing ride

8.20.1 - Omit; Reference not Required

8.22.3 - Omit; Reference not Required

8.22.3.1 - Omit; Reference not Required

8.22 through 8.26 - Teams are expected to produce adequate structural calculations.

The specific modifications outlined in these sections are not required but are eligible for additional points.

8.27 through 8.30 - Fatigue Calculations not required.

8.33 - Teams are expected to produce adequate structural timber calculations. The specific modifications outlined in this section are not required but are eligible for additional points.

8.36.2 - Omit; Reference not Required

SECTION 9: *Hydraulic Equipment for Amusement Rides and Devices*

9.2.2 - Omit; Out of Scope

9.2.2.1 - Omit; Reference not Required

SECTION 10: *Pneumatic Systems and Components*

No Change

SECTION 11: *Safety Related Control Systems*

No Change

SECTION 12: *Electrical Requirements*

No Change

SECTION 13: *Mechanical Systems and Components*

13.2.1 - Omit

13.6.3.3 - Omit; Backup Rollers Out of Scope

SECTION 14: *Fencing, Guardrails, Gates, Walkways for Amusement Rides and Devices*

Reminder - Riders are likely placed by a human, so fences and gates should not apply

SECTION 15: *Welding*

No Change

SECTION 16: *Fasteners*

No Change

SECTION 17: *Operator Controls*

No Change

SECTION 18: *Documentation*

No Change

SECTION 19: *Coatings*

No Change

SECTION 20: *Fall Protection*

No Change

SECTION 21: *Sanitation/Disinfection*

No Change

SECTION 22: *Keywords*

No Change

REC FIG. 1

